

Final Project Peer Review

On a scale 1-5, rate the participation of each of your teammates:

Teammate	Rate
Xie Xianghui	5
Kurtis Gnapp	5
Benjamin Hernandez	5

Rate and describe your contribution to this project:

Student Name	Rate
Alex Santiago	5

Work Description:

I implemented the code to generate the .h audio tables in **Task #1**, following the initial development by Xie Xianghui, who created the code to generate .wav files replicating the desired song and sound behaviors. These audios were later integrated into Tasks #3 and #4. Additionally, I developed a Python script to verify the correctness and integrity of the generated audio data.

For **Task #3**, I led most of the implementation work, with support from Xie Xianghui. I modified the base Pong game code to enhance gameplay functionality, including updates to the finite state machine (FSM) to eliminate extra plays after a win condition was met. I also added new output signals to trigger audio responses, integrated a 7-segment display for improved user experience, and updated both the SLV_REG logic and the overall Pong IP.

Furthermore, I developed the Vitis application responsible for managing the audio response system. This included processing the audio tables, implementing the logic to trigger audio events, and designing the circuitry needed to drive the buzzer effectively.

Throughout the project, I also provided feedback and support to my teammates on their respective tasks as needed and documented the final report, videos and github with some help from my teammates.